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$$T = m_2 g . \qquad (4)$$

For the just-slipping condition, the net force on block 1 must be zero. Equations (2), (3) and (4) gives

$0 = m_2 g - \mu_s m_1 g \cos\theta - m_1 g \sin\theta$

$m_2 = m_1(\mu_s \cos\theta + \sin\theta)$

12. Consider two objects of masses 5 kg and 20 kg which are initially at rest. A force 100 N is applied on the two objects for 5 second.

a) What is the momentum gained by each object after 5 s.
b) What is the speed gained by each object after 5 s.

Final momentum on each object $\Delta P = F\Delta t = 100\text{X } 5 = 500$ kgms^{-1}

Final speed on the object of mass 5 kg = 500 /5 =100 m s^{-1}

Final speed on the object of mass 20 kg =500/20 =25 m s^{-1}

Note that momentum on each object is the same after 5 seconds but speed is not the same after 5 seconds. The heavier mass acquires lesser speed than the one with lower mass.

13. An object of mass 5 kg is initially at rest on the surface. The surface has coefficient kinetic friction $\mu_k = 0.6$. What initial velocity must be given to the object so that it travels 10 m before coming to rest?

When the object moves on the surface it will experience three forces.

a) Downward gravitational force (mg)
b) Upward normal force (N)
c) Frictional force opposite to the motion of the object.

Since there is no motion along the vertical direction, magnitude of normal force is equivalent to the magnitude of gravitational force.

$$N = mg$$

Applying Newton's second law along the x direction

$$m\vec{a} = -\mu_k mg\hat{i}$$

The acceleration is $\vec{a} = -\mu_k mg\hat{i}$

Note that the acceleration is along the x direction since the frictional force acts along the negative x direction.

$$\text{Or} \quad a = -\mu_k g$$

Note that the acceleration is uniform during the entire motion. We can use Newton's kinematic equation to find the final velocity.

$$\text{Along the } x \text{ direction } v^2 = u^2 + 2as$$

Here v = final velocity and u = initial velocity to be given to travel a distance s.

In this problem s = 10 m

Since the particle comes to rest, the final velocity v = 0

$$0 = u^2 - 2\mu_k gs$$

$$u = \sqrt{2\mu_k gs}$$

$$u = \sqrt{2 \times 0.6 \times 9.8 \times 10} = 10.8\ ms^{-1}$$